LGG Brain Tumor Segmentation with U-Net

Project Overview

In this project, the dense pixel-level prediction for brain tumor segmentation is done using the LGG MRI Segmentation Dataset published by Mateusz Buda on Kaggle [1]. The goal is to segment low-grade glioma tumors from 2D FLAIR MRI slices, with the resulting mask being a binary mask for each and every pixel in the image as opposed to a single label or scalar value. The pipeline is intended to be clean, reproducible and fully configured by a single configuration file to ensure that any change in a hyperparameter is uniformly applied across the entire pipeline.

Dataset and Preprocessing

This dataset contains brain MR images together with manual FLAIR abnormality segmentation masks [2]. The MRI slices are from 110 patients with a total of 3,929 slices. The slices are 256x256 3-color RGB TIFF files, and there is also a binary mask file for each slice, identifying the region of the tumor. The 3 channels are pre-contrast, FLAIR, and post-contrast imaging, giving the model a little more information than a single grayscale channel.

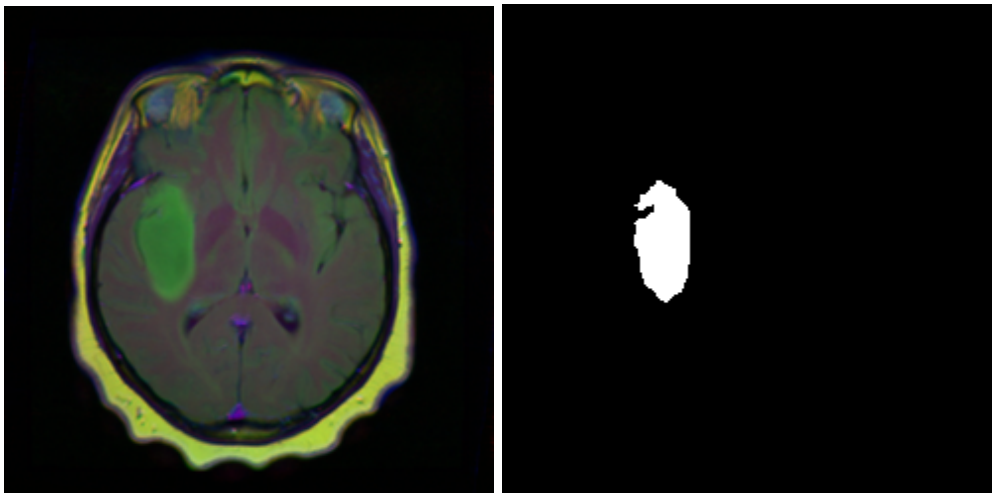


Figure 1: Dataset Sample and Mask

Figure 1. A representative sample from LGG MRI Segmentation Dataset. The brain MRI slice, binary tumor mask, and mask applied to the brain MRI are presented in each row. The overlay shows that the tumor is contained within normal brain tissue.

The main preprocessing choice was that of patient splitting versus slice splitting. The model is able to memorize the patient's anatomy when patient slices are present in both training and test sets, thus the score reported is meaningless. They were then divided into 88 patients for training, 11 patients for validation, and 11 patients for testing, using a shuffle method in which the slices were randomly distributed.A randomly distributed shuffle method of the 110 patient folders yielded 88 patient folders for training, 11 patient folders for validation, and 11 patient folders for testing (3133, 409 and 387 slices respectively). This division guarantees that the test set is truly unseen patients.

During training, only random horizontal and vertical flips were done for augmentation. These are safe transformations for axial brain MRI as they do not deform the anatomy, and also make the model left-right and top-bottom orientation invariant. Validation and test images were not augmented in order to have a deterministic evaluation.

Model Architecture and Training Strategy

The model is a standard U-Net, which is implemented from scratch [3]. The encoder consists of four down-sampling blocks to successively decrease the spatial resolution and increase the channel depth, and the decoder consists of four up-sampling blocks that restore the spatial resolution. The skip connections between the encoder and decoder are important to preserve fine-grained spatial details, especially for tumor boundary segmentation. It is a model with 3 input channels to 1 output channel (logit map).

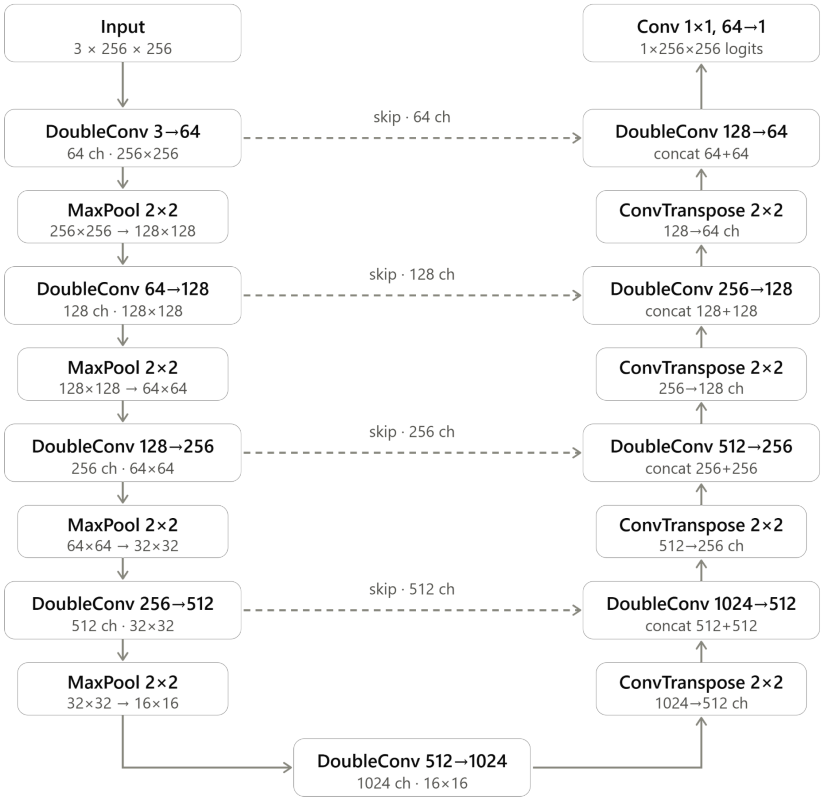


Figure 2: U-Net Architecture Flowchart

Figure 2. Tumor segmentation is accomplished by the U-Net architecture. The contracting path on the left is composed of four down-sampling blocks and results in a reduction of spatial resolution and double channel depth. The spatial resolution is restored in the large path on the right by employing 4 up sampling blocks. Skip connection (grey arrows) feed the encoder features to the decoder features at each resolution. To get to a single-channel logit mask, the output is passed through a last 1x1 convolution.

If the image is a plain binary cross entropy loss, then it will not be able to learn anything useful, since the number of pixels in the image that are tumors is a small minority. The loss function is the Dice loss and binary cross-entropy loss weighted equally. The Dice term directly maximises the amount of overlap between the prediction and ground truth, and the BCE term gives stable gradients early in training. This combination is used frequently and is effective in medical segmentation tasks.

The optimizer is Adam with an initial learning rate of 1e-4 and a weight decay of 1e-5. A ReduceLROnPlateau scheduler reduces the learning rate by half if the validation Dice score doesn't improve over 5 epochs. The model was trained for 100 epochs, using a batch size of 16.

Results

The 11 test patients were held out, and were evaluated with the checkpoint that performed the best on the validation Dice during training. On the 387 test slices, the model reached a Dice coefficient of 0.8073, an IoU of 0.7764, a precision of 0.9293, and a recall of 0.8241.

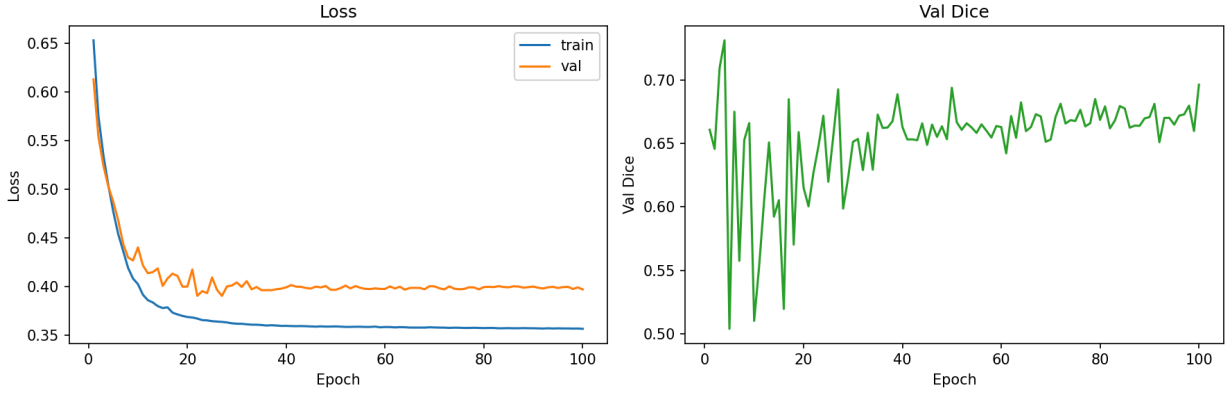


Figure 3: Training Curves

Figure 3. Training and validation loss curves (left) and validation Dice scores (right) with respect to epochs. There is a gradual drop in the training loss from ~0.65 to ~0.35, and the validation loss levels off around epoch 15 and stays around 0.40. The size of the validation set is very small, which results in the validation Dice ranging from about 0.50 to 0.73.

The precision/recall numbers show a meaningful trade-off. The accuracy is about 93 percent: If the model says the pixel is a tumor, it is nearly always right, so there are very few false positives. A recall is an error rate of about 82 percent, which is in the regions of the tumor outline and usually in smaller tumors. This compromise is important in the clinic as the loss of tumor tissue

is typically more expensive than mild over-segmenting of healthy tissue. The model is conservative, a good starting point but that may be improved.

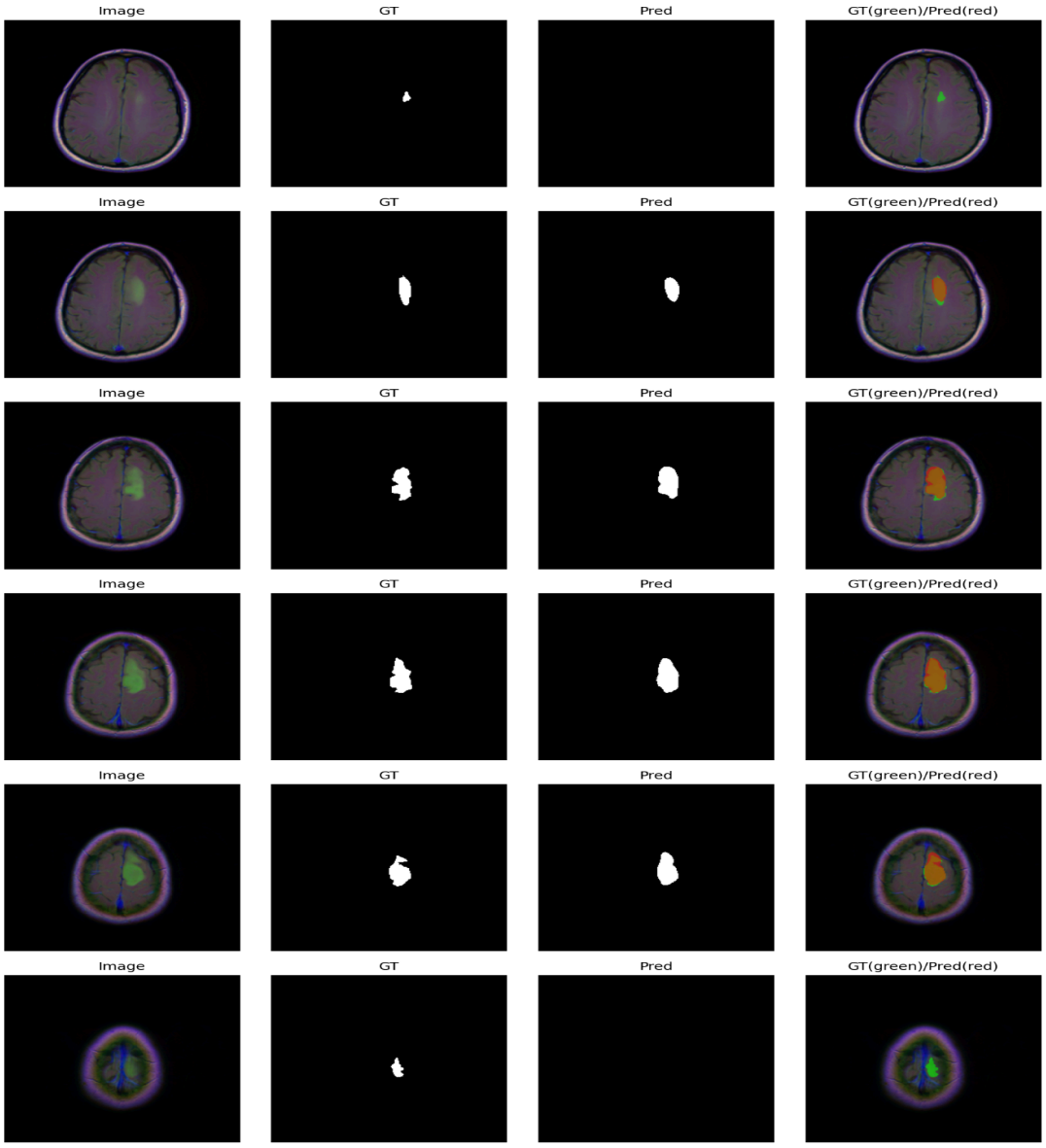


Figure 4: Test Prediction Grid

Figure 4. Qualitative segmentation results are given on a held-out test set of patients. The test slice (column 1), ground truth mask (column 2), model prediction (column 3), and overlaid ground truth and model prediction (column 4) are all displayed for each row. Yellow areas denote correctly segmented regions, green-only areas denote accurate segmentation in regions which do not contain a tumor and red-only areas denote oversegmentation (false positive).

As seen in the training curves, the validation loss was fairly flat starting from epoch 15 and remained around 0.40 throughout the training of 100 epochs, whereas the training loss was gradually reduced from ~0.65 to ~0.35 over the 100 epochs. The validation Dice was variable, with values varying between about 0.50 and 0.73 throughout the epochs. This volatility is directly related to the small size of the validation set (11 patients). Having only a few patients can mean that one challenging case can make a significant impact on the metric. This noise did not affect the performance of the model in generalizing to the test set and it retained its Dice score of over 0.80.

Conclusion

The first lesson from this project is that Dice loss is essential for imbalanced segmentation. With pure BCE the model predicted empty masks for the first few epochs before it learned to detect tumor pixels at all. Adding Dice loss made the learning signal much more direct.

The second lesson is that patient-level splitting is not optional. Splitting by slice would have inflated the Dice score to near perfection by letting the model memorize the exact brain shape of each training patient. Patient-level splitting gives an honest estimate of how the model would perform on a new patient in a new hospital, which is the only number that matters in practice.

The third lesson is that small datasets are volatile. With only 11 validation patients, the validation metric was noisy enough that the best checkpoint was not necessarily the final epoch. Learning rate scheduling helped, but early stopping and careful checkpoint selection were the real safeguards against overfitting.

References

[1] Buda, M., Saha, A., & Mazurowski, M. A. (2019). Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm. Computers in Biology and Medicine, 109, 218-225. (Dataset available at: https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation)

[2] Zhou, B. (2024). Brain MRI Segmentation [Data set]. IEEE DataPort. https://doi.org/10.21227/pv7k-b062 (This provides the description of the 110 patients from TCIA/TCGA).

[3] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. In Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015 (pp. 234-241). Springer.